

Learning Objectives¹

Students will be able to (1) find partial derivatives, (2) multiply vectors and matrices, (3) compute probability and statistics, (4) visualize data with `matplotlib`, and (5) evaluate baseline classifiers in `sklearn`.

You may find the following resources helpful:

- Andrew Ng's CS229 Course (Stanford)
 - Linear Algebra Review (<http://cs229.stanford.edu/section/cs229-linalg.pdf>)
 - Probability Theory Review (<http://cs229.stanford.edu/section/cs229-prob.pdf>)
- Harvey Mudd College Math Department
 - Calculus Tutorials (<https://www.math.hmc.edu/calculus/>)

1 Partial Derivatives (10 pts)

(a) Consider $y = z \sin(x)e^{3x}$. What is the partial derivative of y with respect to x ?

$$\frac{\partial y}{\partial x} = ? \quad \frac{d}{dx} \downarrow \begin{matrix} \sin(x) e^{3x} \\ \cos(x) e^{3x} + \sin(x) \cdot 3e^{3x} \end{matrix}$$

$$\frac{\partial y}{\partial x} = z [\cos(x) e^{3x} + \sin(x) \cdot 3e^{3x}]$$

$$\boxed{\frac{\partial y}{\partial x} = z e^{3x} [\cos(x) + 3\sin(x)]}$$

(b) Consider $f(x, y) = xy^3e^{2x}$. Find $\frac{\partial f}{\partial x}$.

$$x e^{2x}$$

$$e^{2x} + 2x e^{2x}$$

$$\frac{\partial f}{\partial x} = y^3 (e^{2x} + 2x e^{2x})$$

$$\frac{\partial f}{\partial x} = y^3 e^{2x} [1 + 2x]$$

¹This assignment is adapted from course material by Jessica Wu at Harvey Mudd College.

2 Linear Algebra (10 pts)

(a) Consider the matrix $\mathbf{X}$ and the vectors $\mathbf{y}$ and $\mathbf{z}$ below:

$$\mathbf{X} = \begin{pmatrix} 4 & 3 \\ 2 & 5 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad \mathbf{z} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

2×2 2×1 2×1

i. What is the dot product $\mathbf{y}^T \mathbf{z}$?

$$\mathbf{y}^T = (2 \quad 4)$$

$$\mathbf{y}^T \mathbf{z} = 2 \cdot 1 + 4 \cdot 5 = \boxed{22}$$

ii. What is the product $\mathbf{X}\mathbf{y}$?

$$\mathbf{X}\mathbf{y} = \begin{pmatrix} 4 \cdot 2 + 3 \cdot 4 \\ 2 \cdot 2 + 5 \cdot 4 \end{pmatrix} = \begin{pmatrix} 20 \\ 24 \end{pmatrix}$$

(b) Consider the matrix $\mathbf{X}$ and the vectors $\mathbf{y}$ and $\mathbf{z}$ below:

$$\mathbf{X} = \begin{pmatrix} 7 & 2 & 3 \\ 5 & 8 & 1 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 3 \\ 6 \\ 2 \end{pmatrix} \quad \mathbf{z} = \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$$

i. What is the dot product $\mathbf{y}^T \mathbf{z}$?

$$\mathbf{y}^T = (3 \quad 6 \quad 2)$$

$$\mathbf{y}^T \mathbf{z} = 3 \cdot 4 + 6 \cdot 1 + 2 \cdot 7 = \boxed{32}$$

ii. What is the product $\mathbf{X}\mathbf{y}$?

$$\mathbf{X}\mathbf{y} = \begin{pmatrix} 3 \cdot 7 + 6 \cdot 2 + 3 \cdot 2 \\ 5 \cdot 3 + 8 \cdot 6 + 1 \cdot 2 \end{pmatrix} = \begin{pmatrix} 39 \\ 65 \end{pmatrix}$$

3 Probability and Statistics (10 pts)

Consider a sample of data S obtained by flipping a coin X , where 0 denotes the coin turned up heads, and 1 denotes that it turned up tails: $S = \{0, 0, 1, 1, 0\}$.

- (a) What is the sample mean for this data?

$$\bar{x} = \frac{0+0+1+1+0}{5} = \frac{2}{5} = \boxed{0.4}$$

- (b) What is the sample variance?

$$s^2 = \frac{(0-0.4)^2 + (0-0.4)^2 + (0-0.4)^2 + (1-0.4)^2 + (1-0.4)^2 + (0-0.4)^2}{5-1} = \frac{1.2}{4} = \boxed{0.3}$$

- (c) What is the probability of observing this data assuming that a coin with an equal probability of heads and tails was used (i.e., the probability distribution of X is $P(X=1) = P(X=0) = 0.5$)?

$$P(X=0) = 0.5, P(X=1) = 0.5$$

$$P(S) = (0.5)(0.5)(0.5)(0.5)(0.5) = 0.03125$$

- (d) (Optional) Note the probability of this data sample would be greater if the value of $P(X=1)$ was not 0.5 but some other value. Which value maximizes the probability of sample S ?

$$P(X=1) = \frac{2}{5} = \boxed{0.4}$$

- (e) Given the following joint distribution between X and Y , what is $P(X=T|Y=b)$?

$P(X,Y)$		Y		
		a	b	c
X	T	0.2	0.1	0.2
	F	0.05	0.15	0.3

$$P(X=T|Y=b) = \frac{P(X=T, Y=b)}{P(Y=b)} = \frac{0.1}{0.1+0.15} = \frac{0.1}{0.25} = \boxed{0.4}$$

Programming

GitHub Classroom

We will use GitHub Classroom to submit the coding part of assignments. Go to this link to fork the starter code: <https://lmu.postcommit.ai/c/cmsi-5350-fall-2026/assignments/69>

4 GenAI (5 pt)

Read through the rest of the assignment.

- Write down at least one use case for LLM-based AI assistant (e.g., ChatGPT, Gemini, Claude, etc). Example questions include explaining a function from a library that you're not familiar with, providing sample code for how to use the function, or debugging your code.
- Try the question with at least two models from <https://arena.ai/text/side-by-side>. Reflect on the output. Which tools/models did you use? Which model gave you the preferred answer and why? Were they accurate? Helpful? Did they increase your understanding or make you faster at writing code?

Use case: I used an LLM to learn how to use `matplotlib.pyplot.scatter()` to plot Age versus Fare for the Titanic dataset, with the color of each point representing whether the passenger survived.

I compared Gemini 3.6 Flash and Claude Sonnet 4.5. Both models accurately explained the main parameters of `plt.scatter()` and gave useful code examples. Claude began displaying its response about three seconds sooner, but Gemini finished generating its full response first. I preferred Gemini's answer because it was more straightforward and gave a clearer step-by-step explanation of how each part of the scatter plot worked. Claude was also helpful, but it included multiple approaches and extra sample data that I did not really need. Overall, both models improved my understanding and helped me work faster, but Gemini was more efficient for this specific question.

5 Titanic (65 pts)

The sinking of the RMS Titanic² is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the Titanic sank after colliding with an iceberg, killing 1502 out of 2224 passengers and crew. This sensational tragedy shocked the international community and led to better safety regulations for ships.

One of the reasons that the shipwreck led to such loss of life was that there were not enough lifeboats for the passengers and crew. Although there was some element of luck involved in surviving the sinking, some groups of people were more likely to survive than others, such as women, children, and the upper-class.

In this problem, you will complete the analysis of what sorts of people were likely to survive. In particular, you will apply the tools of machine learning to predict which passengers survived the tragedy.

code and data

- code : `titanic.py`
- data : `titanic_train.csv`, `titanic_test.csv`

documentation

- Metrics:
http://scikit-learn.org/stable/modules/generated/sklearn.metrics.accuracy_score.html
-

Note that any portions of the code that you must modify have been indicated with `TODO`. Do not change any code outside of these blocks.

5.1 Visualization (35 pts)

One of the first things to do before trying any formal machine learning technique is to dive into the data. This can include looking for funny values in the data, looking for outliers, looking at the range of feature values, what features seem important, etc.

- (a) Run the code (already provided for you) to make histograms for each feature, separating the examples by class (e.g. survival). This should produce seven plots, one for each feature, and each plot should have two overlapping histograms, with the color of the histogram indicating the class. For each feature, what trends do you observe in the data?

The clearest trends are in passenger class, sex, and fare. First-class passengers had more survivors relative to non-survivors, while third-class passengers had many more non-survivors. Female passengers survived much more often than male passengers, and passengers who survived also tended to have higher fares. Age shows a weaker trend, with survivors being slightly younger overall. For SibSp, which is the number of siblings/spouses aboard, and Parch, which is the number of parents/children aboard, the two survival groups overlap more, so there is not as strong of a pattern. Most passengers embarked from Southampton, while passengers from Cherbourg appear to have a higher proportion of survivors.

²This assignment is adapted from the Kaggle Titanic competition, available at <https://www.kaggle.com/c/titanic>. Some parts of the problem are copied verbatim from Kaggle. Passengers with missing values for any feature have been removed. Also, the categorical feature Sex has been mapped to 'female': 0, 'male': 1 and Embarked to 'Cherbourg': 0, 'Queenstown': 1, 'Southampton': 2. As described in the Kaggle data description, survival has been mapped to 0 = No and 1 = Yes. The ticket class, pclass, is 1 for 1st, 2 for 2nd, and 3 for 3rd. Age is in years, sibsp indicates the number of siblings/spouses aboard, parch, is the number of parents/children aboard, and fare is how much the passenger paid.

- (b) We are often also interested in understanding the interactions between features. Of course, such a task is complicated as (1) visualizing interactions between m out of n features requires $\binom{n}{m}$ plots, and (2) as we live in a 3D-world, we have trouble interpreting plots in high dimensions. Therefore, we will focus here on pairwise interactions.

Make a scatter plot³ of **Age** versus **Fare**, where each point represents an example (e.g. passenger) and the color of the point indicates the class (e.g. survival). In the textbox below, briefly describe any trends you observe.

There does not appear to be a strong relationship between age and survival because survivors and non-survivors are spread across most age ranges. However, passengers who paid higher fares appear more likely to have survived, while many non-survivors are concentrated at lower fares.

5.2 Evaluation (30 pts)

Before trying out any classifier, it is often useful to establish a *baseline*. For this exercise, you will evaluate some simple classifiers on the data. We will model our classifiers after those in `scikit-learn`⁴.

We have implemented one baseline classifier, `MajorityVoteClassifier`, that always predicts the majority class from the training set. Read through the `MajorityVoteClassifier` and its usage and make sure you understand how it works.

Your goal is to implement and evaluate another baseline classifier, `RandomClassifier`, that predicts a target class according to the distribution of classes in the training data set. For example, if 60% of the examples in the training set have `Survived = 0` and 40% have `Survived = 1`, then, when applied to a test set, `RandomClassifier` should randomly predict 60% of the examples as `Survived = 0` and 40% as `Survived = 1`.

Implement the missing portions of `RandomClassifier` according to the provided specifications. Then train your `RandomClassifier` on the entire training data set, and evaluate its training error. If you implemented everything correctly, you should have an error of 0.485.

³You may find the pyplot API at http://matplotlib.org/api/pyplot_api.html useful.

⁴Almost all of the model techniques in `scikit-learn` share a few common named functions, once they are initialized. These are `some-model-name.fit(...)`, `some-model-name.predict(...)`, and `some-model-name.score(...)`. You can always find out more about them in the documentation for each model.

What to turn in

☐ PostCommit

- ☐ Review all modified files for quality, and add your name to `README.md` and all source files.
- ☐ Submit your code to your cloned GitHub repo for this assignment using the following commands. For more details, see [GitHub's documentation](#).

```
git add .  
git commit -m "adding files for homework 1"  
git push
```

☐ Gradescope

- ☐ Finish the survey questions below..
- ☐ Upload your answers to (1) the written questions, (2) survey questions, and (3) any plots/figures to [Gradescope](#).
- ☐ Tag the pages to match each question in Gradescope.

1. What is your GitHub username?

aisaiahmaguadog

2. Number of hours spent working on this assignment:

It took me 4 hours to finish the assignment.

3. What you hope to get out of this course:

A good baseline on the fundamentals of what machine learning is and how to utilize it in my future endeavors.

4. In 1-2 sentences, please describe a positive change you would like to see in the world, and how Machine Learning could help you to achieve that goal:

I want to design hardware in products that can make people's lives easier, and machine learning is a big part of creating technology that can be adaptable.

5. Your previous experience (if any) in machine learning and/or deep learning:

N/A

6. (Optional) Feel free to let me know what you liked/disliked about this homework, what you learned, etc:

I like the organization and having some review before diving into the new topics. This course builds upon material from many different classes, so having some time to remember everything helps a lot.